

System Driven Hardware Design

-24 GHz CW-Radar (Doppler) Modul (IPM-165)-

Version 1.1

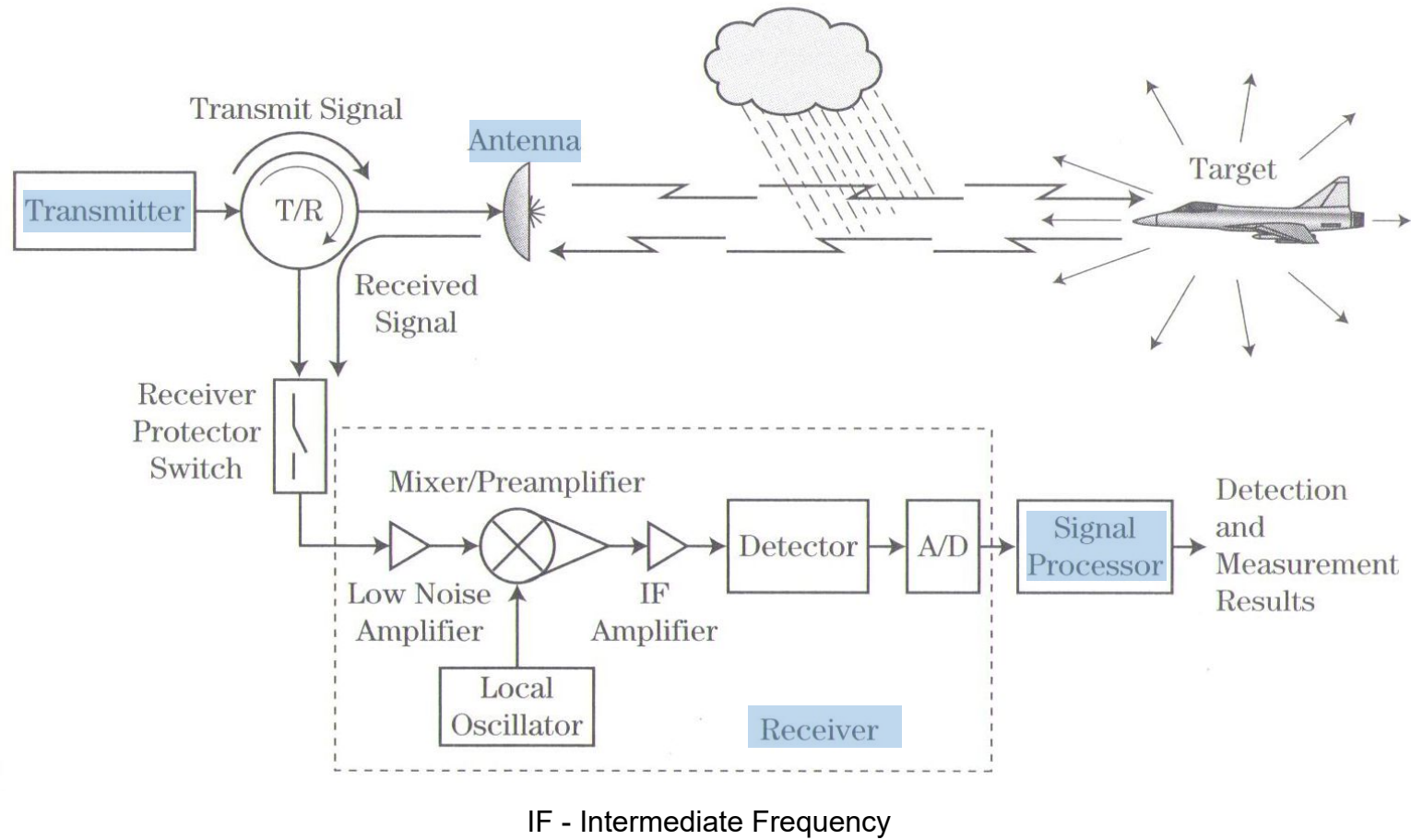
International Master of Science

Prof. Dr.-Ing. Stephan Bannwarth

Understanding the used technique to implement a Doppler shift radar system.
Getting an overall understanding of the radar system and get aware of restrictions from other parts of the system.
Deriving the LF-Amplifier Specification from the datasheet and application note.

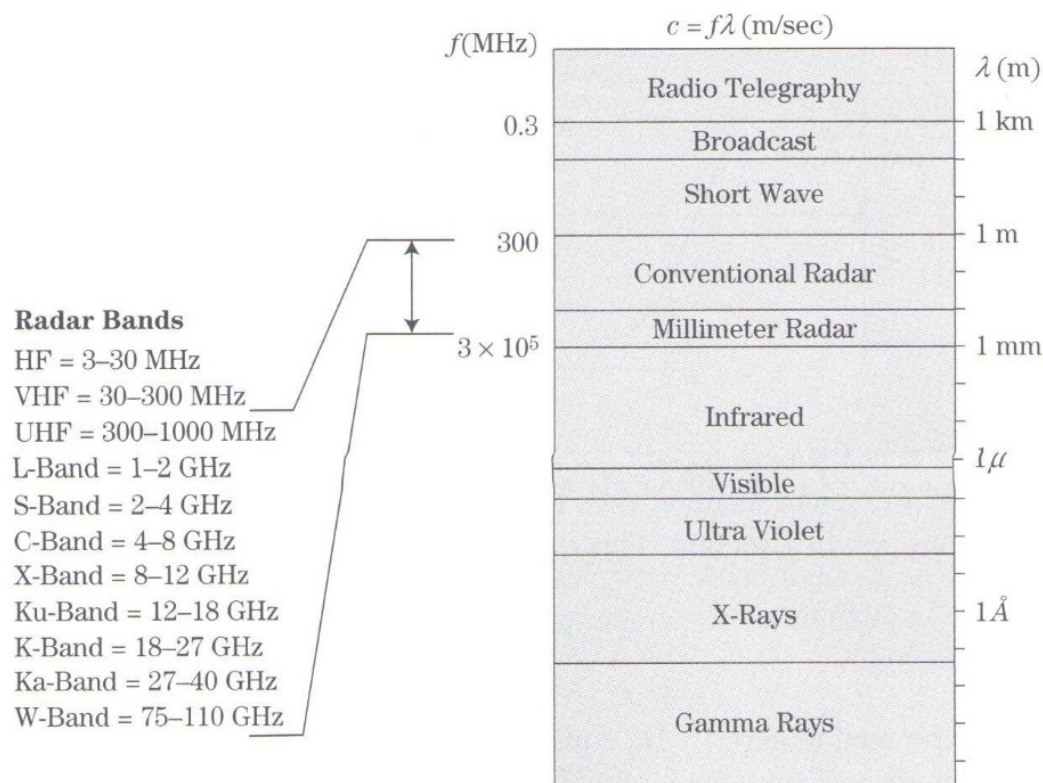
Agenda

- o Radar Fundamentals
- o 24 GHz CW-Radar (Doppler) Modul (IPM-165)
 - Expected Doppler Frequency
 - RF Characteristics
 - LF Characteristics



"Principles of Modern Radar – Basic Principles", SciTech, Richards, Scheer, Hom

Electromagnetic wave types

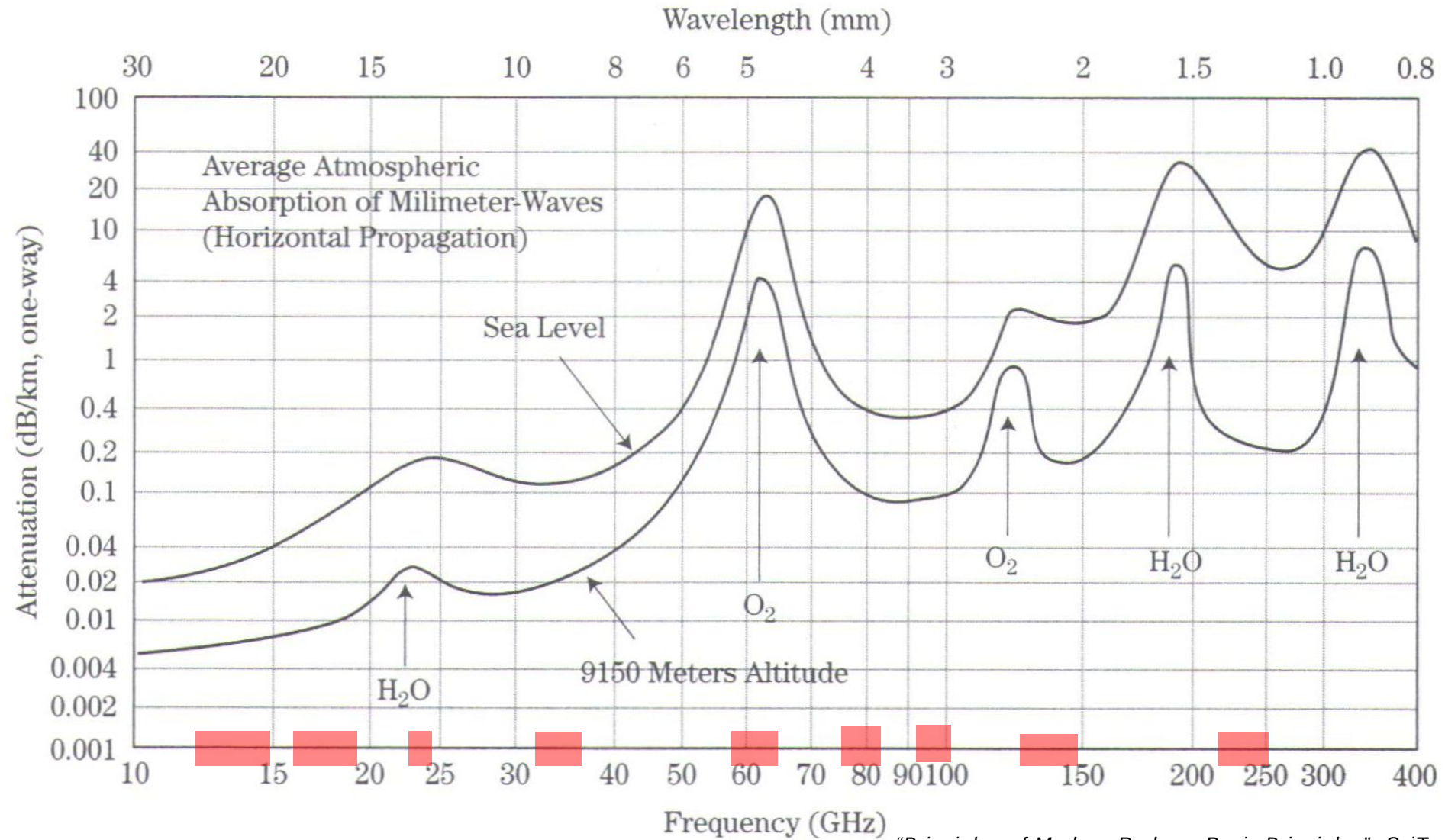


Radio Frequency and Radar Bands

Band	Frequency Range	ITU Radar Freq.
High frequency (HF)	3–30 MHz	
Very high frequency (VHF)	30–300 MHz	138–144 MHz 216–225 MHz
Ultra high frequency (UHF)	300 MHz–1 GHz	420–450 MHz 890–942 MHz
L	1–2 GHz	1.215–1.400 GHz
S	2–4 GHz	2.3–2.5 GHz 2.7–3.7 GHz
C	4–8 GHz	5.250–5.925 GHz
X	8–12 GHz	8.500–10.680 GHz
Ku (“under” K-band)	12–18 GHz	13.4–14.0 GHz 15.7–17.7 GHz
K	18–27 GHz	24.05–24.25 GHz 24.65–24.75 GHz
Ka (“above” K-band)	27–40 GHz	33.4–36.0 GHz
V	40–75 GHz	59.0–64.0 GHz
W	75–110 GHz	76.0–81.0 GHz 92.0–100.0 GHz
mm	100–300 GHz	126.0–142.0 GHz 144.0–149.0 GHz 231.0–235.0 GHz 238.0–248.0 GHz

Major of radars work between: 300MHz to 35GHz (automobile: 76GHz and 110 GHz)

“Principles of Modern Radar – Basic Principles”, SciTech, Richards, Scheer, Hom



ITU - RF

"Principles of Modern Radar – Basic Principles", SciTech, Richards, Scheer, Hom

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Description

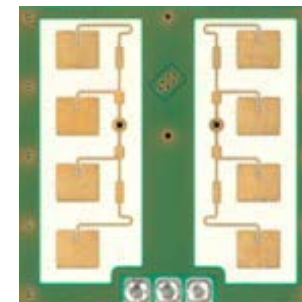
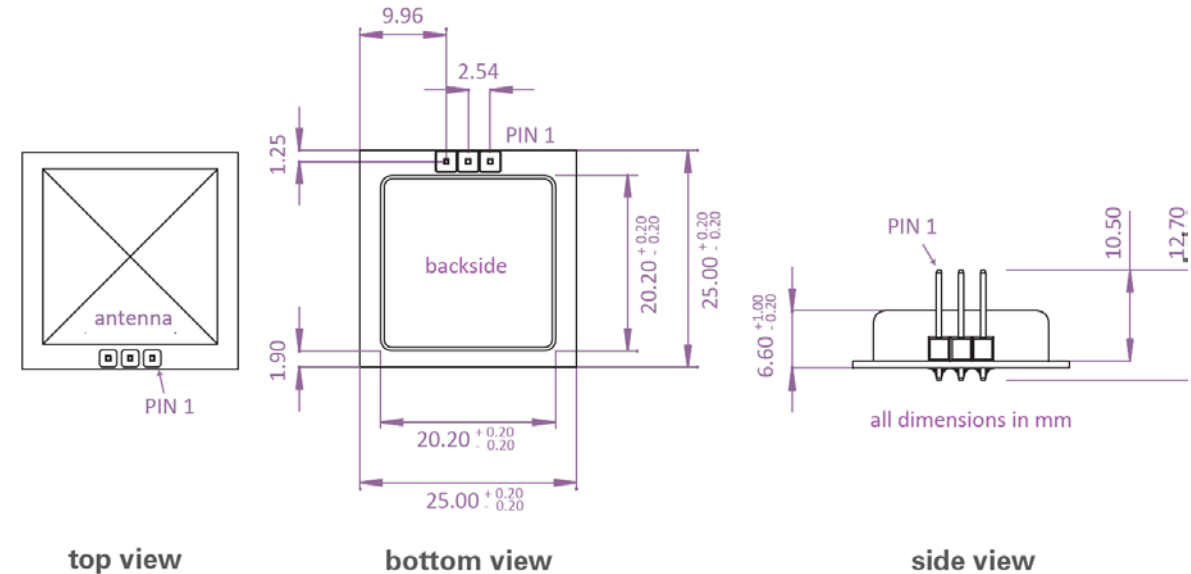
The IPM-165 is a 24GHz Doppler module with an asymmetrical wide beam for detection of moving objects.

Interface

The sensor provides a 2.54mm grid, single row pin header (square pin $\square 0.635\text{mm}$)

PIN #	DESCRIPTION	IN / OUT	COMMENT
1	V_{CC}	input	supply voltage (4.75 - 5.25V)
2	IF1	output	signal I(nphase)
3	GND	input	analog ground

IPM 165 Mechanical Outlines



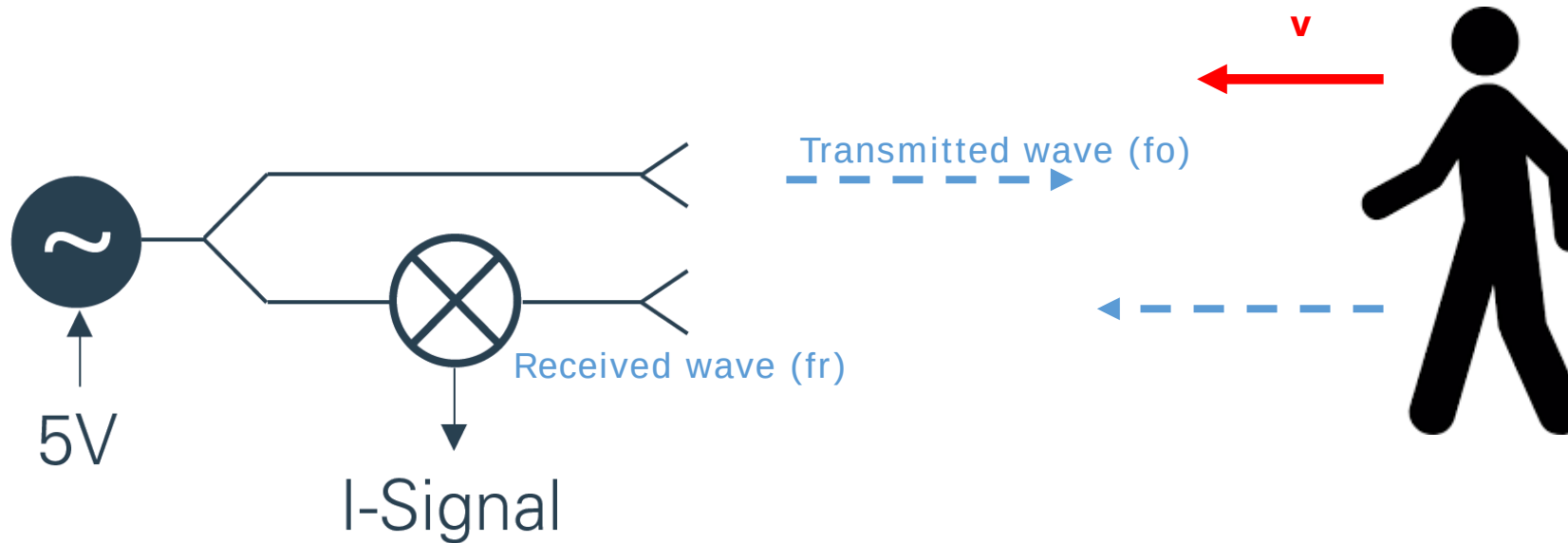


Fig 2: Block diagram of the IPM 165
(Innosent Application Note III)

Principal

The Transmitted wave consist ideally of a fixed frequency f_o . The frequency f_r of the received wave is shifted by the Doppler Effect:

$$f_r = f_o \left(1 \pm \frac{2v}{c} \right)$$

Question

- Where is the factor 2 coming from?
- What is the I-Signal and what is the Mixer doing?

In mixer we obtain difference frequency ($f_r - f_o$) -> gives doppler frequency

InnoSent – IPM-165 Application Note III

I-signal is Inphase signalling and IPM 165 has only I-signal and no Q-signal. That is the reason why we can only detect velocity of the target and not the direction of object movement (towards or away from us)

Object	Speed v / (km/h)		Doppler Frequency f_d (Hz)
Hand	2,52	0.7 meter/hr	112.7 Hz
Walking	5	1.389 meter/hr	223.63 Hz
Running	12	3.33 meter/hr	536.13 Hz
Car	30	8.33 meter/hr	1.341 KHz
	50	13.889 meter/hr	2.236 KHz
	120	33.33 meter/hr	5.366 KHz

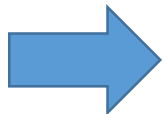
$$f_d = f_o \frac{2 v}{c}$$

$$c = 3,0 \cdot 10^8 \text{ m/s}$$

$$f_o = 24,15 \text{ GHz}$$

$$f_o/c = 80.5 \text{ m}$$

$$1 \text{ kmph} = 5/18 \text{ meter/hour}$$



Hertz per 1 m/s:
Expected I-Signal Frequency Span:

Agenda

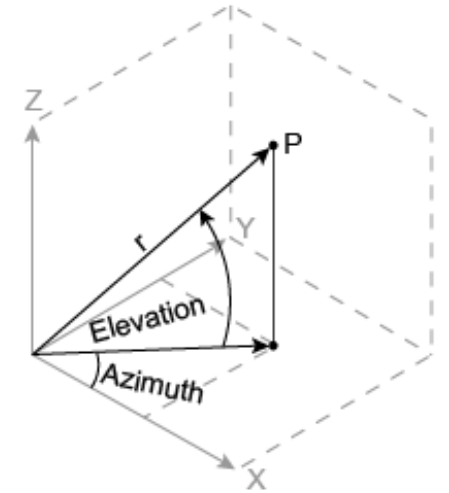
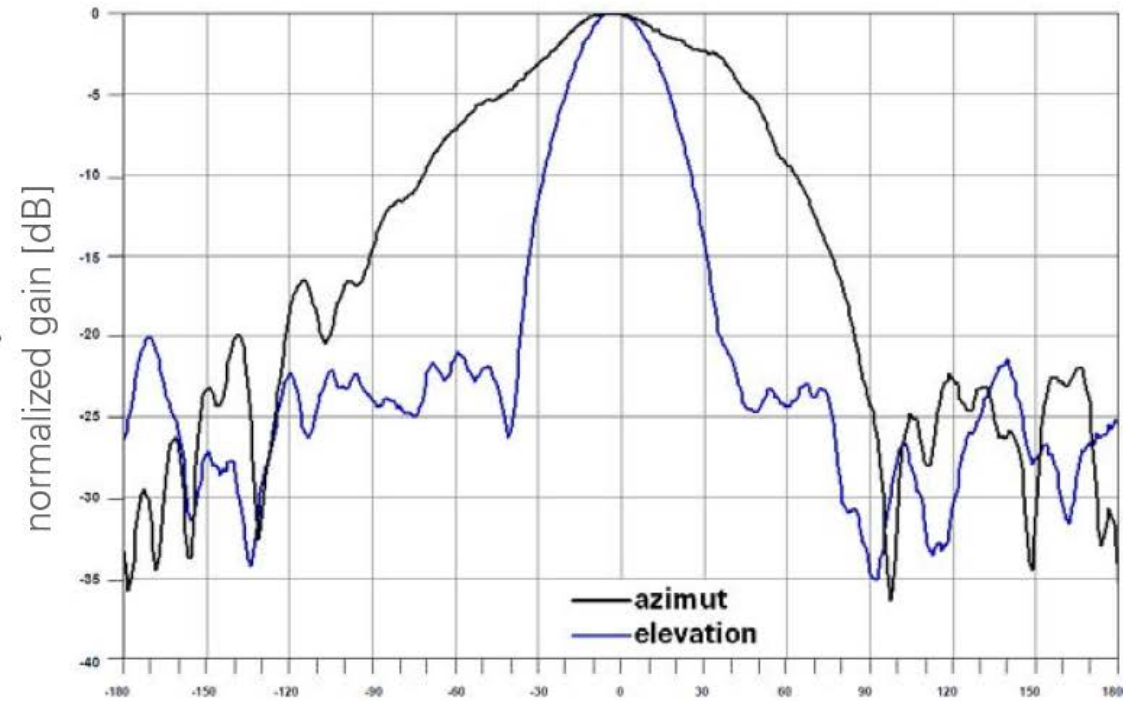
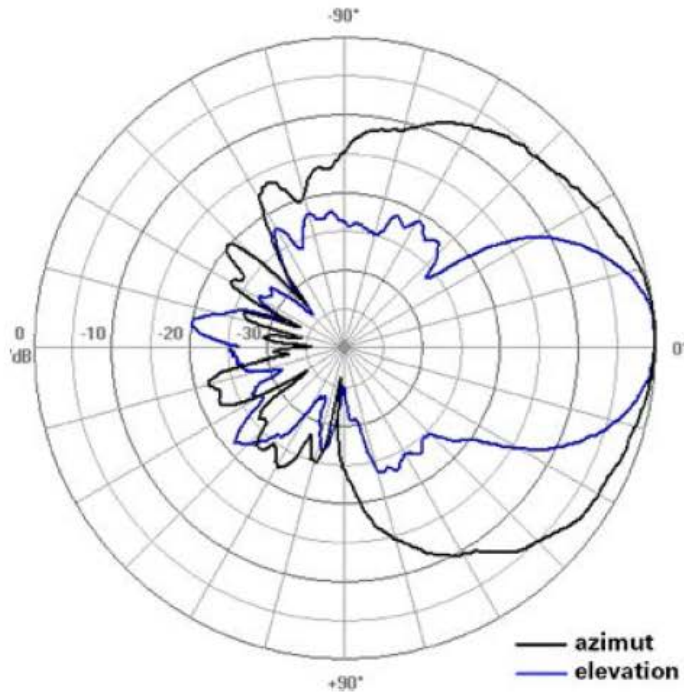
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Oscillator

transmit frequency ¹	@ 25°C	f_{standard}	24.050		24.250	GHz
	US-frequency band @ 25°C	f_{F}	24.075		24.175	GHz
	UK-frequency band @ 25°C	f_{UK}	24.150		24.250	GHz
output power		P_{out}		16	20	dBm
temperature drift		Δf		-1		MHz/°C

$$\lambda = 1,24 \text{ cm}$$

TX / RX-ANTENNA PATTERN:

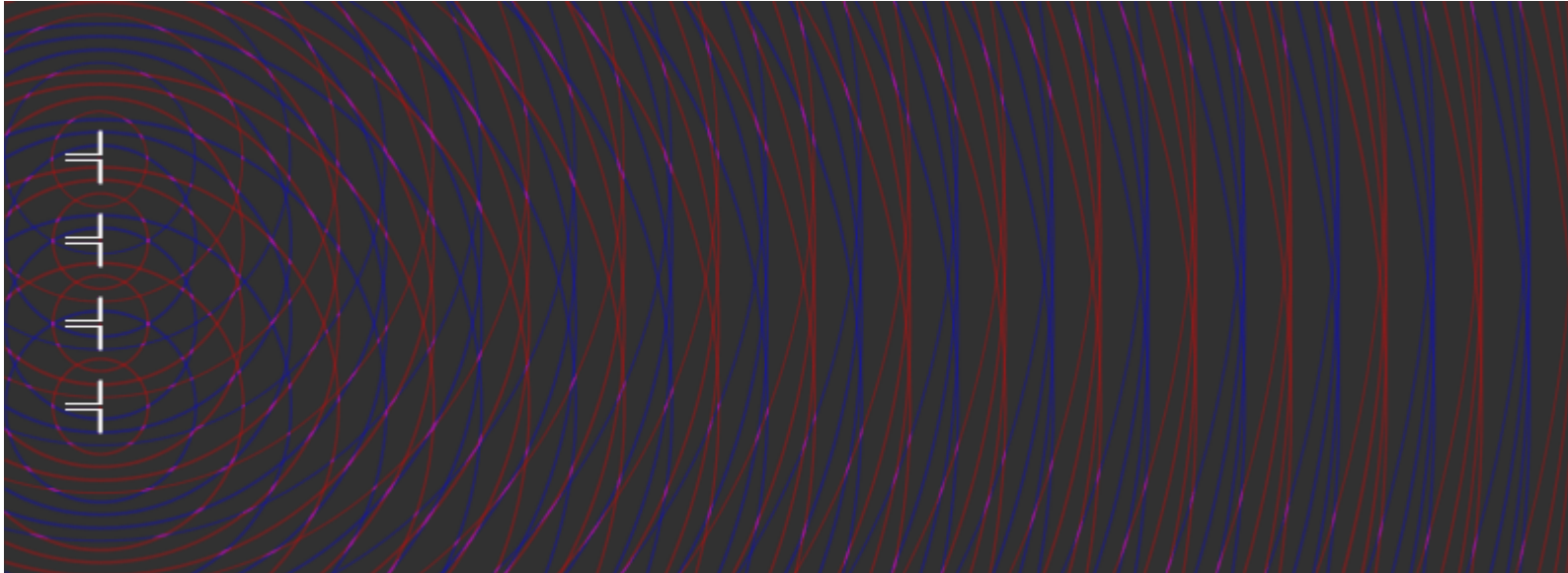


InnoSent – IPM-165 Datasheet

Near and Far Field Region

Far field region is known as Fraunhofer region.
Near field region is sometimes referred to as Fresnel region.

We do not use near field area for doppler frequency and we use far field because of interference



Antennas in the size of multiple wavelengths

$$r_{far} = \frac{2 D^2}{\lambda}$$

Near field is the region within a radius $r \ll \lambda$
Far field is the region for which $r \gg \lambda$

D: dimension of the antenna
 λ : Wavelength
r: minimal distance of the far field

In our case: $r_f = 6,67\text{cm}$

<https://www.radartutorial.eu/06.antennas/an60.en.html>

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Receiver

IF-output		voltage offset	-300		300	mV
Signal level ²		category A	563		855	mV _{P-P}
		category B	856		1177	mV _{P-P}
		category C	1178		1819	mV _{P-P}
Noise level ²		R			116	mV

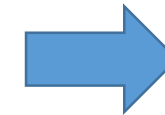
Power supply

supply voltage		V _{cc}	4.75	5.00	5.25	V
supply current		I _{cc}	18	30	40	mA

INTERFACE

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1	V _{cc}	input	supply voltage (4.75 - 5.25V)
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What is missing?

- Any word about the output impedance of the I-Signal.
- Any word about the expected signal level of the I-Signal.

InnoSent – IPM165 Datasheet

Excerpts from the Application Notes

A few more remarks regarding the IPM-165

Low signal amplitude (300uV)

Internal output resistance is middle ohmic (I-Signal)

LF-amplifier Circuitry

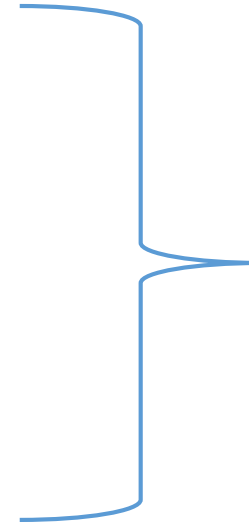
High input impedance

AC-coupling suggested, because of DC offset at the mixer

70-80dB voltage gain

Mixer output bandwidth is at least 100MHz

Nominal transmit frequency is 24.125 GHz



Decision

We stick to our starting point of **A=300** and **V_{in,min} = 300uV**. We then calculate the maximum input amplitude, allowed by the ADC input range.

Note

Later the gain might be changed.

Literature

- *“Principles of Modern Radar – Basic Principles”*, SciTech, Richards, Scheer, Hom
- InnoSent Datasheets of the IPM-165
- InnoSent Application Notes III (*IPM-165 – a universal Low Cost K-Band Transceiver for Motion Detection in various Applications*)